NATIONAL INSTITUTE OF TECHNOLOGY SIKKIM

Customer Segmentation and E-Commerce Mining

Data Warehouse and Mining Laboratory (AI14201 / CS14211)

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Revised report: **27 October 2026**. This revision supersedes the original report and reports the results of the saved notebooks.

Abstract

This project builds a SQLite session warehouse and evaluates future spending class, future session inactivity and next-period revenue. A fixed customer holdout and training-only cross-validation separate model selection from final evaluation. Descriptive mining uses log-transformed RFM and multivariate session anomaly detection.

The audit exposed arbitrary row-order tie breaking in spending labels, unsupported historical claims and a same-period revenue target. This revision uses declared future-revenue bands, a fully observed 90-day inactivity horizon, and a next-90-day revenue target. These changes make old and new scores non-comparable.

The resulting predictive evidence is weak. Spending macro-F1 is only slightly above a majority baseline, churn ranking is weak, and lower revenue MAE than a mean baseline is explained by zero-heavy outcomes. A broader training-only nested-validation search did not establish a stronger model. The contribution is an auditable pipeline and an honest assessment of its limits.

Scope	Evidence
Data	25,000 sessions; 14,214 catalogue rows; 7,875 eligible customers
Feature cutoff	2024-09-30
Outcome horizon end	2024-12-29
Reporting rule	Every reported model score comes from a saved notebook output.

1. Data architecture and ETL

The repository retains two notebooks: notebook 01 validates and loads sources; notebook 02 performs modelling and descriptive mining. The supplied workbook is retained but is not joined because a shared customer identifier is undocumented.

Source	Verified use
Ecommerce.csv	25,000 session records with dates, revenue, device, payment, channel, category and interaction attributes.
Fashion Dataset v2.csv	14,214 catalogue rows. Category-level price and rating aggregates provide metadata proxies.
Online Shopping Behavior Dataset.xlsx	Retained unchanged; not used for modelling or unsupported customer-level joins.

Star-schema tables

Table	Grain / role
fact_sessions	One row per session; all 25,000 sessions are retained.
dim_customer	Source customer identifiers; no fabricated slowly changing history.
dim_geography	Undocumented source location codes, without invented Indian states.
dim_festival	Manual 2024 festival windows; first matching window wins on overlap.
dim_fashion_style	Category price/rating metadata and proxy coverage.
mart_multimodal_customers	One row per eligible customer; past predictors and separately named future outcomes.

Time window and outcomes

Past features use visits on or before 30 September 2024. Outcomes use visits after that cutoff through 29 December 2024 inclusive, a 90-day horizon. Customers first seen after the cutoff do not enter the modelling population. The final source date outside the horizon is excluded from labels.

Spending band	Future revenue rule	Customers
Low	Exactly zero	6778
Medium	Positive and at most 1,000	394
High	Above 1,000	703

These fixed bands are illustrative, not business-validated thresholds; the source currency is undocumented. Identical future revenue receives identical labels, unlike the old row-order ranking rule.

2. Feature design and evaluation protocol

Customer split

A seed-42 stratified split is fixed before any initial model fit: 6,300 training and 1,575 test customers, with no customer overlap. The same customer partition is reused for spending, churn and revenue. Spending stratification does not imply separate churn stratification of the outer holdout.

Partition	Low	Medium	High	Total
Train	5,423	315	562	6,300
Test	1,355	79	141	1,575

Predictor groups

Group	Features and rationale
RFM	Recency, distinct session frequency and past monetary sum. Past spending is allowed because the target is future revenue.
Behaviour	Session count; average/median unit price; quantity; device, channel, payment and category shares; festival share; tenure; visit intervals; purchase/cart shares and discounts.
Catalogue proxies	Observed-category-weighted catalogue price, average rating, normalised price and coverage. No images are processed.
Further validation	Recent 30/60/90/180-day counts and spending, trends, purchase timing, last-session attributes and calendar summaries. The expanded set has 99 predictors.

Leakage controls

Future monetary, future session count, churn, class labels and customer IDs are excluded from predictors. Features use past visits only. A perturbation of future revenue, unit price and device data must leave every past feature unchanged. Source keys, nonnegative values, dimension coverage and SQLite integrity are checked.

Initial model selection uses five-fold training CV. Median imputation, scaling and optional SMOTE occur inside each fold. XGBoost sample weights are recomputed on the fold training labels; LightGBM class weights are fold-local. Randomised searches maximise macro-F1 for spending. No held-out threshold, feature or model selection occurs.

After all initial supervised selections are locked, each prespecified estimator is evaluated in one final test stage. Later figures and error inspection use cached predictions. Further development uses nested validation on original training customers only, with hashes proving that original test outputs remain unchanged.

Selection budgets and compute

Each spending tree family/imbalance approach samples four configurations, and logistic regression samples three. Searches use one CV job and two native CPU threads. Temperature monitoring is external; pause at 75 degrees C and remain below 85 degrees C. No GPU training is used.

3. Spending prediction and ablation

Training CV selected XGBoost with SMOTE and RFM + behavioural + catalogue-proxy features. Each ablation selects its own estimator using the same candidate families and search budget; the comparison therefore measures modelling recipes rather than a causal proxy effect.

Recipe / baseline	Accuracy	Balanced accuracy	Macro-F1	Low recall	Medium recall	High recall
Majority baseline	0.860317	0.333333	0.308305	1.000000	0.000000	0.000000
Logistic regression baseline	0.860317	0.333333	0.308305	1.000000	0.000000	0.000000
RFM	0.860317	0.333333	0.308305	1.000000	0.000000	0.000000
RFM + behaviour	0.719365	0.310229	0.311916	0.825830	0.012658	0.092199
RFM + behaviour + proxies	0.707302	0.319593	0.320110	0.808856	0.050633	0.099291

Scores above are final held-out results. The selected model raises macro-F1 slightly above the majority baseline while losing accuracy and balanced accuracy. It recalls few Medium or High customers; the evidence does not support strong targeting performance.

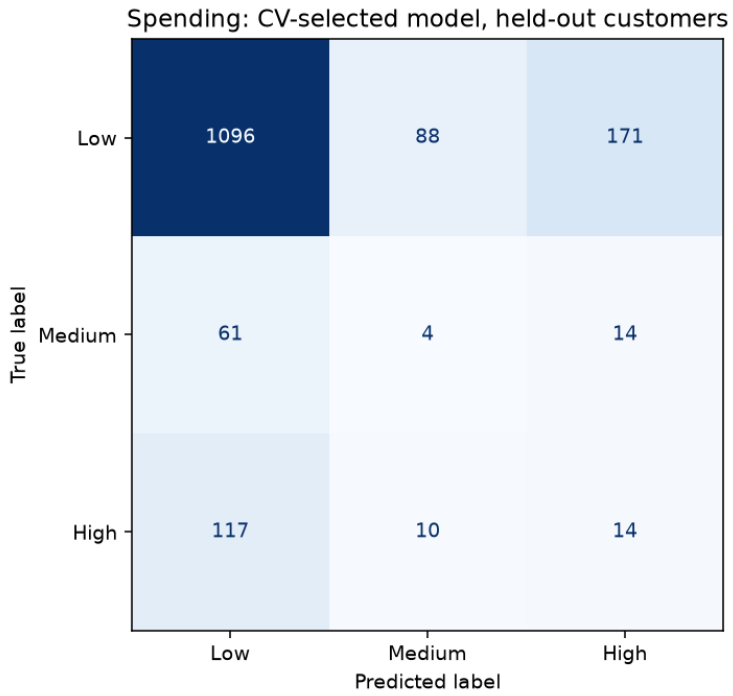


Figure 1. Locked spending model on held-out customers; rows are actual classes and columns are predictions.

The 14 Medium customers predicted High are an error cohort, not validated aspirational prospects. No in-sample marketing target list is produced. The unsupported fixed +6% visual improvement claim is replaced by the measured ablation above.

Source: outputs/spending_test_metrics.csv and outputs/spending_confusion_matrix.csv.

4. Future inactivity and revenue prediction

Churn: no session in the next 90 days

The old churn rule, recency above 90 days, is replaced with absence of sessions in the future horizon. Logistic regression and random forest are compared by training-CV average precision. Logistic regression wins that selection. The original test decision threshold remains 0.5; PR-AUC here means average precision.

Model / baseline	ROC-AUC	PR-AUC (AP)	Churn F1	Accuracy	Macro-F1
Prior baseline	0.500000	0.514286	0.000000	0.485714	0.326923
Logistic regression	0.471185	0.492620	0.500311	0.490159	0.489948

The model ranks worse than the constant-score baseline on this holdout. The prior baseline uses training prevalence; at 0.5 it predicts the active class, so its zero churn F1 is a weak F1 reference. The later nested analysis adds an always-churn reference. Session absence is not confirmed account closure.

CLV proxy: next-90-day revenue

Past features predict future monetary revenue, rather than explaining the same-period monetary column. Linear regression and raw/log1p-target histogram gradient boosting are compared by training-CV MAE on the original revenue scale. The log-target model is selected; inverse predictions use expm1 without a smearing correction.

Model / baseline	R-squared	MAE
Mean baseline	-0.000077	454.987265
Linear regression	-0.011268	457.160529
Gradient boosting / raw target	-0.009499	455.653103
Gradient boosting / log1p	-0.095658	261.689656

The selected model improves MAE relative to a mean predictor, but its negative R-squared indicates weak squared-error forecasting. With mostly zero outcomes, a near-zero estimate can improve MAE without learning useful revenue separation. The nested analysis tests this against a median constant. This target is not discounted lifetime value.

Sources: outputs/churn_test_metrics.csv and outputs/clv_test_metrics.csv.

5. Additional training-only validation

After the original holdout was opened, a requested second round explored stronger models without reopening it. Three outer folds assess a procedure with five inner folds. All customers in this extension belong to the original training partition. Estimates remain exploratory because algorithm development has already used this population; a new future period is still required.

Spending: richer features and decision weighting

Compare logistic regression, Extra Trees, histogram boosting, CatBoost at two depths, and the original fixed XGBoost recipe. Inner out-of-fold probabilities choose the candidate and class multipliers by macro-F1. Each locked choice is evaluated once on its outer fold. Multipliers are decision weights, not calibrated probabilities.

Procedure / reference	Mean macro-F1	Fold std	Mean accuracy	Mean balanced accuracy
Majority baseline	0.308397	0.000053	0.860794	0.333333
Nested selected procedure	0.316990	0.001800	0.785556	0.325481
Original fixed recipe	0.330954	0.015348	0.691111	0.336218

The broader selection does not outperform the original recipe on these outer folds. That recipe was previously chosen using the full training set, so it is an exploratory reference, not an independently selected nested competitor. Inner full-training selection is optimistic and does not justify promoting a new model.

Churn: ranking and threshold selection

Procedure / reference	Mean ROC-AUC	Mean PR-AUC (AP)	Mean churn F1
Always churn reference	0.500000	0.499841	0.666526
Nested selected procedure	0.505205	0.500749	0.666384

Ranking remains near chance. Threshold tuning produces approximately the always-churn reference F1, rather than meaningful customer separation.

Revenue: test the median reference

Procedure / reference	Mean R-squared	Mean MAE
Mean baseline	-0.000307	461.970898
Median baseline	-0.088537	267.732483
Nested selected procedure	-0.088537	267.732483

The training-median constant wins inner selection in every outer fold. Extra modelling did not beat this simple reference. These validation means must not be subtracted from held-out scores to claim an improvement.

6. Log-RFM customer segmentation

Recency, frequency and monetary are log1p-transformed and standardised before K-Means. K is selected from 2 through 6 using the highest silhouette on a fixed sample of up to 2,000 customers. Davies-Bouldin and inertia are diagnostics. Gaussian-mixture BIC is a separate model-family diagnostic, not a K-Means score.

K	Silhouette	Davies-Bouldin	GMM BIC
2.000000	0.389507	1.161009	10266.767977
3.000000	0.373690	1.072131	4982.567194
4.000000	0.377325	1.021710	-8631.769322
5.000000	0.391887	0.957688	-8620.474077
6.000000	0.407408	0.914251	-22682.417514

Selected segment profiles

ID	Customers	Mean recency	Mean frequency	Mean monetary	Median monetary
0	1360	75.21	3.84	2515.59	1953.03
1	1656	148.24	1.00	0.03	0.00
2	929	15.09	2.58	0.00	0.00
3	1912	104.31	2.62	0.00	0.00
4	716	10.50	3.23	2467.49	2042.78
5	1302	119.93	1.62	1915.66	1418.84

K=6 wins silhouette within the tested range and also has the lowest tested GMM BIC. Because the optimum is at the search boundary, it does not establish six natural or stable customer types. IDs describe past activity; labels such as VIP or sleepers are not validated business outcomes.

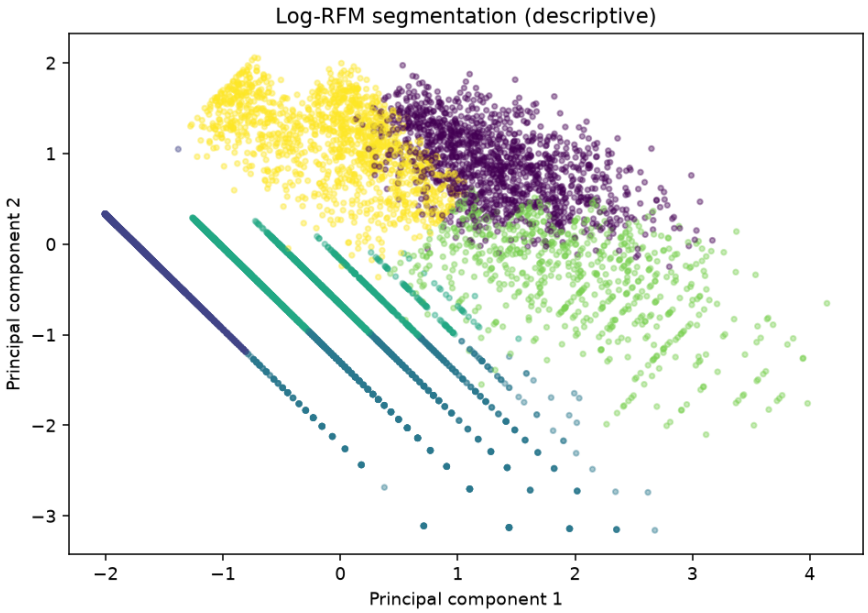


Figure 2. PCA projection of the descriptive log-RFM segments; all eligible customers are included.

7. Explainability and model interpretation

SHAP explains the locked initial spending classifier on a deterministic sample of 150 training customers. Imputation and scaling use the fitted training pipeline. The explanation never selects features from test outcomes. Tree models use TreeExplainer; linear models use LinearExplainer if selected.

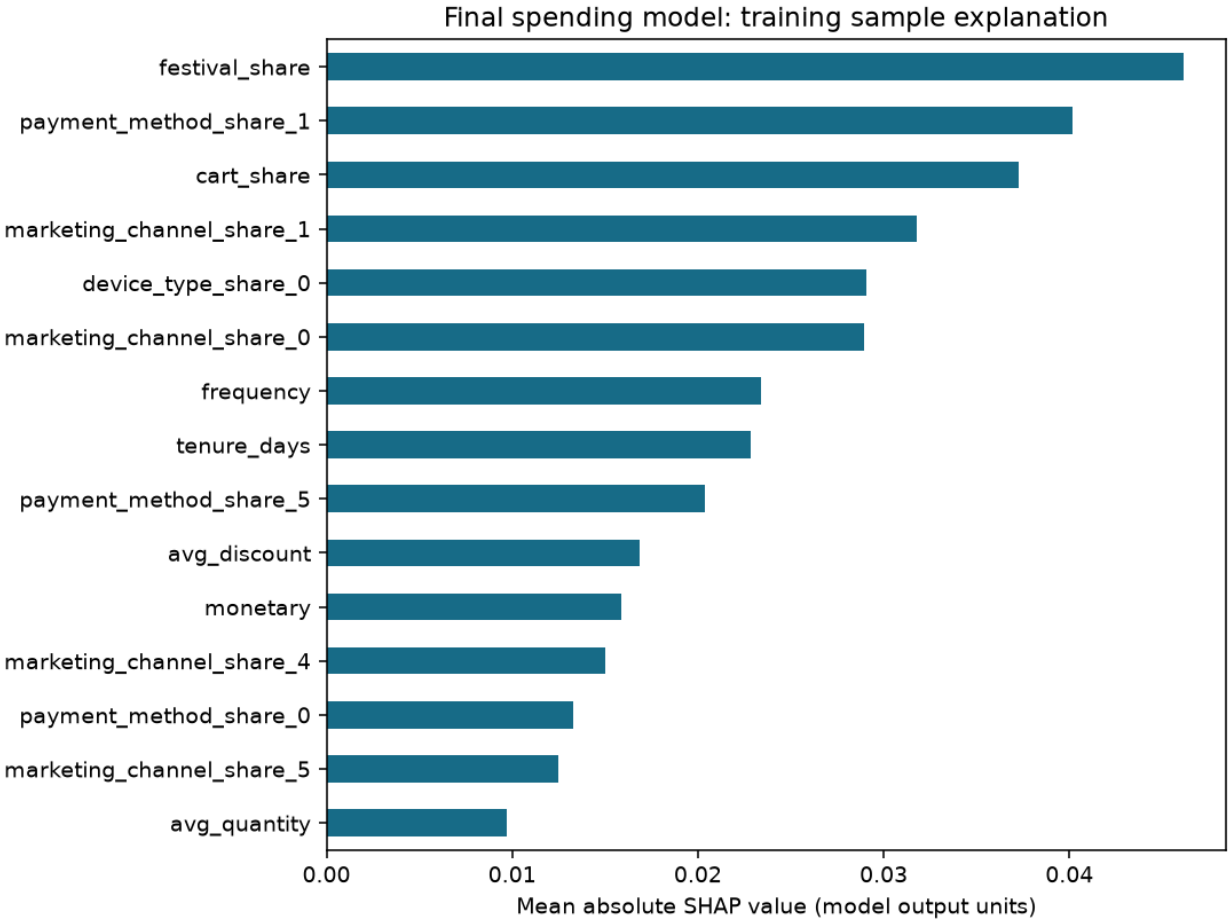


Figure 3. Mean absolute SHAP attribution across sampled training customers and output classes.

The displayed predictors describe how this particular fitted model uses its inputs. A high attribution does not show causation, predictive reliability, campaign lift or customer intent. The weak held-out metrics remain the main evidence of utility.

Catalogue price, average rating and normalised price are metadata aggregates, weighted by past category preferences. The category mapping is assumed, product joins are unverified, and missing categories are not filled with future session statistics. No ResNet-50 embeddings, PyTorch image pipeline or verified visual-feature uplift is present.

Sources: outputs/spending_shap_importance.csv and the saved SHAP figure from notebook 02.

8. Multivariate session anomalies and limitations

Isolation Forest uses log1p revenue, unit price, quantity, pages viewed, time on site and discount percentage. It is fitted descriptively on all 25,000 sessions, separately from predictive modelling. Configured contamination is 0.01; 250 sessions are flagged.

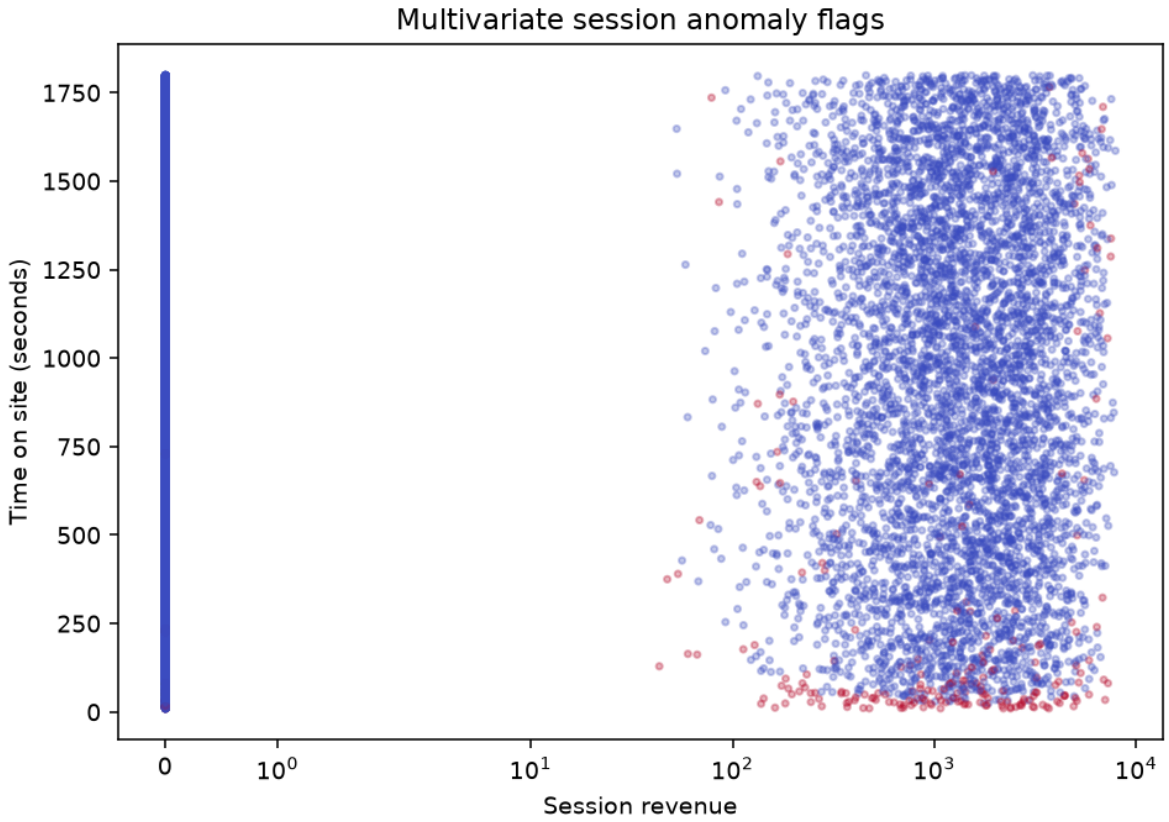


Figure 4. Session revenue versus time on site; red points are multivariate anomaly flags, not verified fraud.

Flags are statistical unusualness under a specified detector. The chosen contamination controls the approximate flag rate; it is not detection accuracy. There are no confirmed fraud or bot labels and no claim of validated fraud detection.

Practical limits

One cutoff cannot establish temporal generalisation. Revenue bands are illustrative, the currency is undocumented and zero outcomes dominate spending and revenue. Catalogue dates and cross-source identity are unknown. Geography codes have no verified state mapping. Festival windows are manually curated and do not support causal promotion-effect claims.

The warehouse is an educational SQLite implementation. Segment stability, model calibration, economic value and deployment performance have not been established. New verified customer/product links, stronger historical signals and a genuinely untouched future period are needed before making operational recommendations.

9. Corrected claims and conclusion

Historical claim / issue	Current evidence
Approximately 88% churn accuracy	Unsupported. The old saved notebook printed 0.63 accuracy; the new future-inactivity results are reported with ranking metrics and baselines.
ResNet-50 visual embeddings	Absent. The implementation uses catalogue category-level price/rating proxies.
Fixed +6% visual-feature gain	Unsupported. A measured feature-group ablation replaces the claim; it does not establish causal or statistically significant uplift.
Old macro-F1 approximately 0.454	The saved XGBoost macro-F1 is 0.455329; 0.454055 is balanced accuracy.
14,000 catalogue products	The source contains 14,214 rows. This is not a verified cross-source product join.
Balanced future-spending classes	Row-order tie breaking split identical zero revenue across classes. Declared bands replace that arbitrary target.
Aspirational marketing targets	Medium-to-High misclassification is an error cohort, not demonstrated marketing opportunity.
Geographic, festival or fraud conclusions	No invented state mapping, causal festival lift or validated fraud detection is claimed.

Historical versus current evaluation

Metric	Historical saved output	Current held-out output
Spending accuracy / macro-F1	0.540317 / 0.455329	0.707302 / 0.320110
Churn accuracy / macro-F1	0.63 / 0.60 (saved precision)	0.490159 / 0.489948
Revenue R-squared	0.0861	-0.095658

These rows are historical context, not a controlled before/after experiment. Spending labels, churn labels, revenue targets and customer splitting changed. A higher spending accuracy cannot be advertised as a genuine improvement across those different tasks.

Conclusion

The revision delivers a reproducible warehouse, explicit future outcomes, fold-local training, auditable baselines and saved evidence. Additional model complexity did not establish a stronger predictive model. Honest limitations and a reproducible negative result are preferable to unsupported performance or business claims.

10. Reproduction and evidence index

The report is generated from saved notebook exports. Building it does not refit a model, reopen the test set or modify saved scores. The original report is preserved under local_notes/original_docs/ and excluded from Git. Its SHA-256 is recorded in the report manifest.

Run the analysis

Use Python 3.12 on Windows. Create and activate a virtual environment, then install the pinned analysis environment. Run commands from the repository root.

```
python -m pip install -r requirements-lock.txt

python scripts/run_notebooks.py

python scripts/verify_results.py

python -m pip install -r requirements-report.txt

python scripts/build_report.py

python scripts/verify_report.py
```

The runner uses the current Python executable, executes both notebooks sequentially and saves outputs. Monitor temperature externally; thermal.pause pauses between fits. Interrupt immediately if needed during a fit. Do not run concurrent training jobs.

Evidence index

Artifact	Role
01_Data_Warehouse_ETL_updated.ipynb	Source validation, temporal mart, warehouse and leakage checks.
02_Multimodal_Data_Mining_updated.ipynb	Initial training selection, final test, explanations, descriptive mining and separate nested validation.
outputs/model_summary.json	Authoritative initial held-out model metrics.
outputs/additional_validation_summary.json	Training-only nested estimates and unchanged holdout hashes.
outputs/legacy_audit.json	Historical saved-output audit with source commit.
outputs/customer_split.csv	Fixed, customer-disjoint train/test partition.
outputs/*_test_metrics.csv	Per-model held-out scores; cached predictions support verification.
Project_Report.md / scripts/build_report.py	Readable report source and reproducible PDF builder.
outputs/report_manifest.json	Source/output hashes, numerical reporting checks and report build versions.

Project repository: https://github.com/ChaitanyaKash/CustomerSegmentation. References in this report are repository artifacts, not claims about unverified external dataset provenance. Slides were absent from the repository; their location remains unresolved and no updated slide deck is claimed.